

FIITJEE VADODARA CENTRE

PHYSICS , MATHEMATICS

PHASE – 5

DATE: 30/07/2024

Time Allotted: 3 Hours

Maximum Marks: 182

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section 1** is Physics, **Section 2** is Chemistry and **Section 3** is Mathematics.
4. All the section can be filled in **PART-A** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Marking Scheme for Only One Part.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –1 In all other cases.
 For example, if **(A), (C) and (D)** are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only **(A) and (D)** will result in **+2 marks**; and darkening **(A) and (B)** will result in **–1 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-14)** – Contains seven (07) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-A (15-18)** - This section contains Two paragraphs. Based on each paragraph, there are Two multiple choice questions. Each question has only one correct answer and carries **+3 marks** for the correct answer and **-1 marks** for wrong answer.

SECTION – 1: PHYSICS

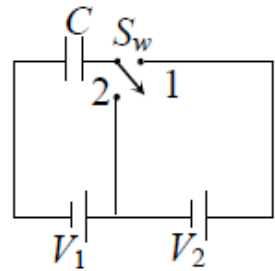
PART A: (One or More Than One Options Correct Type)

This section contains 7 multiple choice type questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

1. The switch S_w is shifted from position 1 to position 2 as shown in figure. Heat generated in the circuit is (Neglect work done by external agent)

(A) independent of V_1
(C) depends on V_1

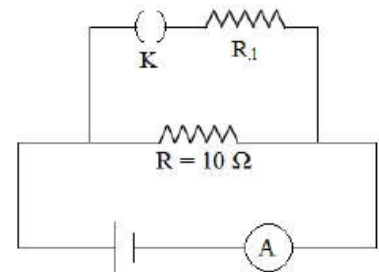
(B) independent of V_2
(D) depends on V_2



2. A parallel plate capacitor of capacitance C has its plates at a separation 'd'. Its capacitance can be increased by:
- (A) Inserting a dielectric slab of thickness $d/2$ between the capacitor plates
(B) Inserting a conductor slab of thickness $d/2$ between the capacitor plates
(C) Decreasing the separation between the plates of capacitor
(D) Increasing the separation between the plates of capacitor

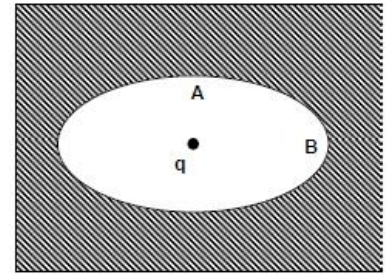
3. The internal resistance of the cell shown in figure is negligible. On closing the key, the ammeter reading changes from 0.25 A to $5/12\text{ A}$, then:

(A) $R_1 = 10\Omega$
(B) $R_1 = 15\Omega$
(C) The power drawn from the cell increases
(D) The current through R decreases by 40%



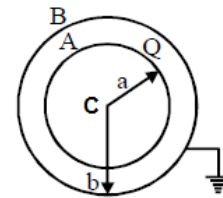
space for rough work

4. In a uniform electric field, the equi-potential surfaces must be:
(A) plane surfaces
(B) normal to the direction of electric field every-where
(C) surfaces with equal potential difference are separated by equal distance
(D) have decreasing potentials in the direction of the electric field
5. An ellipsoidal cavity is carved within a perfect conductor. A positive charge q is placed at the centre of the cavity. The points A and B are on the cavity surface as shown in the figure. Then
(A) Electric field near A in the cavity is equal to electric field near B in the cavity
(B) Charge density at A is equal to charge density at B
(C) Potential at A equals to potential at B
(D) The total electric flux through the surface of the cavity is q/ϵ_0



space for rough work

6. A conducting sphere A of radius a , with charge Q , is placed concentrically inside a conducting shell B of radius b . B is earthed. C is the common centre of the A and B



(A) The field at a distance r from C, where

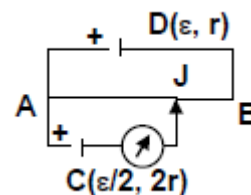
$$a \leq r \leq b \text{ is } \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

(B) The potential at a distance r from C, where $a \leq r \leq b$, is $\frac{1}{4\pi\epsilon_0} \frac{Q}{r}$

(C) The potential difference between A and B is $\frac{1}{4\pi\epsilon_0} Q \left(\frac{1}{a} - \frac{1}{b} \right)$

(D) The potential at a distance r from C, where $a \leq r \leq b$, $\frac{1}{4\pi\epsilon_0} Q \left(\frac{1}{r} - \frac{1}{b} \right)$

7. In the potentiometer arrangement shown, the driving cell D has emf ϵ and internal resistance r . The cell C, whose emf is to be measured, has emf $\epsilon/2$ and internal resistance $2r$. The potentiometer wire is 100-cm long. If balance is obtained, the length $AJ = \ell$



(A) $\ell = 50\text{cm}$

(B) $\ell > 50\text{cm}$

(C) Balance will be obtained only if resistance of AB is $> r$.

(D) Balance cannot be obtained.

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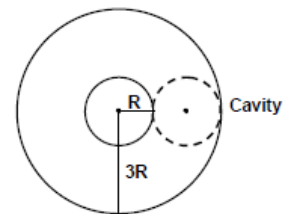
PART A : (Only One Option Correct Type)

This section contains **7 Only One One choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE option is correct**.

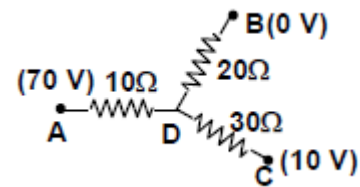
8. n identical cells, each of emf ε and internal resistance r , are joined in series to form a closed circuit. The potential difference across any one cell is
- (A) zero (B) ε (C) $\frac{\varepsilon}{N}$ (D) $\frac{n-1}{n}\varepsilon$
9. Two identical pendulums, A and B are suspended from the same point. The bobs are given positive charges, with A having more charge than B. They diverge and reach equilibrium, with A and B making angles θ_1 and θ_2 with the vertical respectively.
- (A) $\theta_1 > \theta_2$ (B) $\theta_1 < \theta_2$
 (C) $\theta_1 = \theta_2$ (B) The tension in A is greater than that in B
-

space for rough work

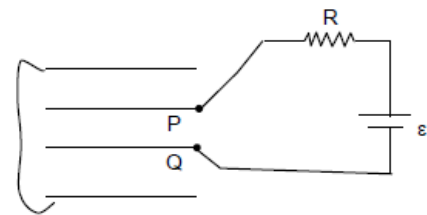
10. A thick shell with inner radius R and outer radius $3R$ has a uniform charge density $\sigma \text{ C/m}^3$. It has a spherical cavity of radius R as shown in the figure. The electric field at the centre of the cavity is
- (A) zero (B) $2\sigma R / \epsilon_0$
(C) $3\sigma R / 4\epsilon_0$ (D) $7\sigma R / 12\epsilon_0$



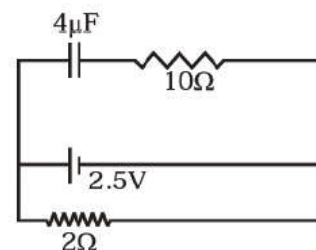
11. In the network shown, points A, B and C are at potentials of 70 V, zero and 10 V respectively.
- (A) Point D is at a potential of 20 V.
(B) The currents in the sections AD, DB, DC are in the ratio 3 : 2 : 1.
(C) The currents in the sections AD, DB, DC are in the ratio 1 : 2 : 3.
(D) The network draws a total power of 400 W.



12. Four identical plates are located in air at equal distance d from each other. The area of each plate is A and the battery EMF is ϵ .
- (A) effective capacitance between P and Q is $\frac{3\epsilon_0 A}{2d}$
(B) in the steady state charge on plate P is $\frac{2\epsilon\epsilon_0 A}{3d}$
(C) Higher the value of R , earlier will be the steady state.
(D) In steady state current flowing through the circuit is ϵ/R



13. Which of the following characteristics of the electrons determines the current in a conductor?
- (A) Drift velocity alone (B) Thermal velocity alone
(C) Both drift velocity and thermal velocity (D) Neither drift velocity nor thermal velocity
14. A capacitor of $4 \mu\text{F}$ is connected as shown in the circuit. The internal resistance of the battery is 0.5Ω . The amount of charge on the capacitor plates will be
- (A) $0 \mu\text{C}$ (B) $4 \mu\text{C}$
(C) $8 \mu\text{C}$ (D) $10 \mu\text{C}$



space for rough work

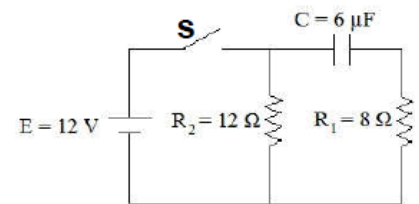
PART A: Comprehension Type (Only One Option Correct)

This section contains **2 Paragraphs**, each describing theory, experiments, data etc. **4 Questions** relate to the **2 Paragraphs** with **2 Questions** on **Each Paragraph**. Each question has **Only One Correct Answer** among the four given options (A), (B), (C) and (D).

Paragraph for Question Nos. 15 to 16

The circuit shown in figure contains an ideal battery E and other elements arranged as shown. The capacitor is initially uncharged and switch S is closed at $t = 0$.

(use $e^2 = 7.4$)



15. The time constant of the circuit is
 (A) $48 \mu\text{s}$ (B) $28.8 \mu\text{s}$ (C) $72 \mu\text{s}$ (D) 120μ
16. The potential difference across the capacitor in volts, after two time constants, is
 (A) 2 (B) 7.6 (C) 10.4 (D) 12

space for rough work

Paragraph for Question Nos. 17 to 18

A charged ball of mass m and positive charge q is projected from the ground with an initial velocity u and angle of projection θ . At the time of projection, a vertically upward electric field 'E' is switched on. When it reaches the maximum height, the vertical electric field is switched off and a horizontal electric field of same magnitude 'E' is switched on in the direction of velocity. Then answer the following question (Assume $mg > qE$)

17. If the time of ascent to the maximum height is t_1 and the time of decent is t_2 then

- (A) $t_1 = t_2$ (B) $t_1 < t_2$ (C) $t_1 > t_2$ (D) None

18. The maximum height reached by the projectile is

- (A) $\frac{u^2 \sin^2 \theta}{2g}$ (B) $\frac{u^2 \sin^2 \theta}{2\left(g - \frac{qE}{M}\right)}$ (C) $\frac{u^2 \sin^2 \theta}{2\left(g + \frac{qE}{M}\right)}$ (D) $\frac{u^2 \sin^2 \theta}{2\left(\frac{qE}{M} - g\right)}$

space for rough work

SECTION - 2: MATHEMATICS

PART-A: (One or More Than One Options Correct Type)

This section contains 7 multiple choice type questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

1. Which of the following are rational number

(A) $\sin\left(\tan^{-1} 3 + \tan^{-1} \frac{1}{3}\right)$ (B) $\cos\left(\frac{\pi}{2} - \sin^{-1} \frac{3}{4}\right)$
 (C) $\log_2\left(\sin\left(\frac{1}{4} \sin^{-1} \frac{\sqrt{63}}{8}\right)\right)$ (D) $\tan\left(\frac{1}{2} \cos^{-1} \frac{\sqrt{5}}{3}\right)$

2. If $\lim_{x \rightarrow \infty} f(x^2) = a$ (a is finite number) then which of the following is/are true

(A) $\lim_{x \rightarrow \infty} x^2 f'(x^2) = 0$ (B) $\lim_{x \rightarrow \infty} x^2 f'(x^2) = 2a$
 (C) $\lim_{x \rightarrow \infty} x^4 f''(x^2) = 0$ (D) $\lim_{x \rightarrow \infty} x^4 f'''(x^2) = a$

3. $f(x) = \frac{3x^2 + ax + a + 1}{x^2 + x - 2}$, then which of the following can be correct?

(A) $\lim_{x \rightarrow 1} f(x)$ exists $\Rightarrow a = -2$ (B) $\lim_{x \rightarrow -2} f(x)$ exists $\Rightarrow a = 13$
 (C) $\lim_{x \rightarrow 1} f(x) = 4/3$ (D) $\lim_{x \rightarrow -2} f(x) = \frac{-1}{3}$

space for rough work

4. If $f(x) = \{x\} |\sin \pi x|$ then (note: $[k]$ and $\{k\}$ denotes the greatest integer and fractional part functions of k respectively)
- (A) Continuous $\forall x \in \mathbb{R}$ (B) Not differentiable at integers
(C) L.H.D at integers is π (D) L.H.D at integers is $-\pi$
5. Let $g(x) = x^4 - 4x^3 + 10$. Compress the graph vertically by a factor of 2 followed by a 180° Rotation across the x-axis, we get a function say $f(x)$ then
- (A) no of real solution of the equation $(f(f(f(x)))) = 0$ is 8
(B) no of real solution of the equation $(f(f(f(x)))) = 0$ is 6
(C) if $f : [-1, 3] \rightarrow [15/2, 17/2]$, then $f^{-1}(-5)$ is 0
(D) range of $f(x)$ is $\left[-\infty, \frac{17}{2}\right]$
6. If the function $f(x)$ and $g(x)$ are defined on $\mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \begin{cases} 0, & x \in \text{rational} \\ x, & x \in \text{Irrational} \end{cases}$
- $g(x) = \begin{cases} 0, & x \in \text{Irrational} \\ x, & x \in \text{rational} \end{cases}$ then $(f - g)(x)$ is
- (A) one – one and onto (B) neither one – one nor onto
(C) one – one but not onto (D) onto but not one-one
7. D is domain and R is range of $f(x) = \sqrt{x-1} + 2\sqrt{3-x}$ then
- (A) $D : [1, 3]$ (B) $D : (-\infty, 1] \cup [3, \infty)$ (C) $R : [1, \sqrt{3}]$ (D) $R : [\sqrt{2}, \sqrt{10}]$

space for rough work

PART-A: (Only One Option Correct Type)

This section contains 7 **Only One choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE option is correct**.

8. The value of $\sum_{r=0}^{\infty} \tan^{-1}\left(\frac{1}{1+r+r^2}\right)$ is
 (A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) $\frac{3\pi}{4}$ (D) None of these
9. Number of values of x which are not in the domain of derivative of the function

$$f(x) = \begin{cases} (x+1)^{3/5} + \pi & ; x < -1 \\ \cos^{-1}(4x^3 - 3x) & ; -1 \leq x < 1 \\ (x-1)^{3/5} & ; 1 \leq x < 2 \end{cases}$$

 (A) 2 (B) 3 (C) 4 (D) 5
10. Let $f(x)$ be a function defined in $[0,5]$ such that $f^2(x) = 1 \forall x \in [0,5]$ and $f(x)$ is discontinuous only at all integers in $[0,5]$ then total number of possible function(s) is (are)
 (A) 100 (B) 99 (C) 162 (D) 194
-

space for rough work

11. Let R be relation from the set $(N \times N)$ to $(N \times N)$ and it is defined by $(a, b) R (c, d) \Rightarrow a + d = b + c$. Then R is
 (A) symmetric only (B) transitive only (C) reflexive only (D) equivalence only
12. Let $f(x) = \begin{cases} x, & -1 \leq x \leq 1 \\ x^2, & 1 < x \leq 2 \end{cases}$ the range of $h^{-1}(x)$ where $h(x) = f \circ f(x)$ is
 (A) $[-1, \sqrt{2}]$ (B) $[-1, 2]$ (C) $[-1, 4]$ (D) $[-2, 2]$
13. Let $f(x)$ be a function whose domain is $[-5, 7]$. Let $g(x) = |2x + 5|$ then domain of $(f \circ g)(x)$ is
 (A) $[-4, 1]$ (B) $[-5, 1]$ (C) $[-6, 1]$ (D) $[-5, 7]$
14. The value of the expression $\left[\frac{18}{35} \right] + \left[\frac{18(2)}{35} \right] + \left[\frac{18(3)}{35} \right] + \dots + \left[\frac{18(33)}{35} \right] + \left[\frac{18(34)}{35} \right]$ is
 (Note : $[k]$ denotes the greatest integer function less than equal to x)
 (A) $(17)^2$ (B) $(18)^2$ (C) $(19)^2$ (D) 195
-

space for rough work

PART-A: Comprehension Type (Only One Option Correct)

This section contains **2 Paragraphs**, each describing theory, experiments, data etc. **4 Questions** relate to the **2 Paragraphs** with **2 Questions** on **Each Paragraph**. Each question has **Only One Correct Answer** among the four given options (A), (B), (C) and (D).

Paragraph for Question Nos. 15 to 16

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$, defined as $f(x) = x^2 e^{-x}$

15. Which of the following is true?
 (A) $f(x)$ has a local maximum at $x = 0$ and a local minimum at $x = 2$
 (B) $f(x)$ has a local minimum at $x = 0$ and a local maximum at $x = 2$
 (C) $\lim_{x \rightarrow +\infty} f(x) = 1$
 (D) None of these
16. Number of solutions of the equation $4x^2 e^{-x} - 1 = 0$ is
 (A) 0 (B) 1 (C) 2 (D) 3

Paragraph for Question Nos. 17 to 18

Let $f(x)$ be a real valued function satisfying $2f(xy) = [f(x)]^y + [f(y)]^x \forall x, y \in \mathbb{R}$ and $f(1) = 3$.

17. $\lim_{x \rightarrow 0} \frac{f(x) + f(-x) - f(f(0)) + 1}{x^2}$ is equal to
 (A) $\ln 3$ (B) $(\ln 3)^2$ (C) $\ln(3^2)$ (D) 0
18. $f(x)$ is a
 (A) monotonically increasing function $\forall x \in \mathbb{R}$
 (B) monotonically decreasing function $\forall x \in \mathbb{R}$
 (C) monotonically increasing function $\forall x \in \mathbb{R} - f(0)$
 (D) monotonically decreasing function $\forall x \in \mathbb{R} - f(0)$

space for rough work

SECTION – 3 : CHEMISTRY

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & C** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

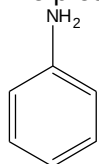
C. Marking Scheme For All Two Part.

- (i) **Part-A (01-04)** – Contains Four (04) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **–1 marks** for wrong answer.
- (ii) **PART–A (05–10)** contains Six (6) Multiple Choice Questions which have One or More Correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –1 In all other cases.
For example, if **(A), (C) and (D)** are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only **(A) and (D)** will result in **+2 marks**; and darkening **(A) and (B)** will result in **–1 marks**, as a wrong option is also darkened.
- (iii) **PART–C (01-08)** contains (08) Numerical based questions with single digit integer as answer, ranging from 0 to 9 (both inclusive) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

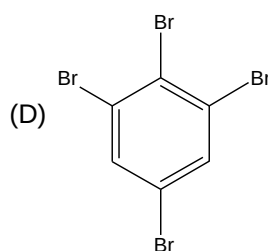
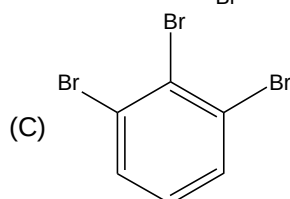
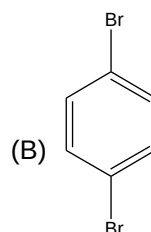
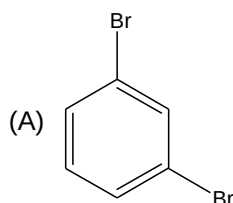
SECTION – II : CHEMISTRY**PART – A: (Single Correct Answer Type)**

This section contains **04 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

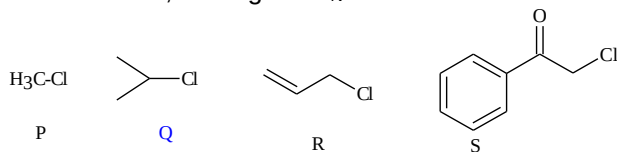
1. The product(s) of the following reaction sequence is (are)



- i) Acetic anhydride/pyridine
ii) KBrO_3/HBr
iii) H_2O^+ , heat
iv) NaNO_2/HCl , 273–278 K
v) Cu/HBr



2. KI in acetone, undergoes $\text{S}_{\text{N}}2$ reaction with each of P, Q, R and S. The rates of the reaction vary as

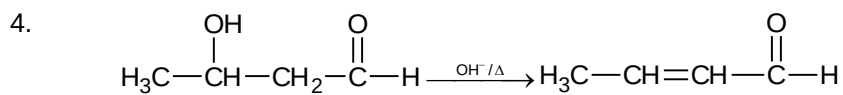


- (A) $\text{P} > \text{Q} > \text{R} > \text{S}$
(C) $\text{P} > \text{R} > \text{Q} > \text{S}$

- (B) $\text{S} > \text{P} > \text{R} > \text{Q}$
(D) $\text{R} > \text{P} > \text{S} > \text{Q}$

Space for rough work

3. Which of the following is not a reducing sugar
- (A) Maltose (B) Galactose
(C) Sucrose (D) Fructose



Expected mechanism is

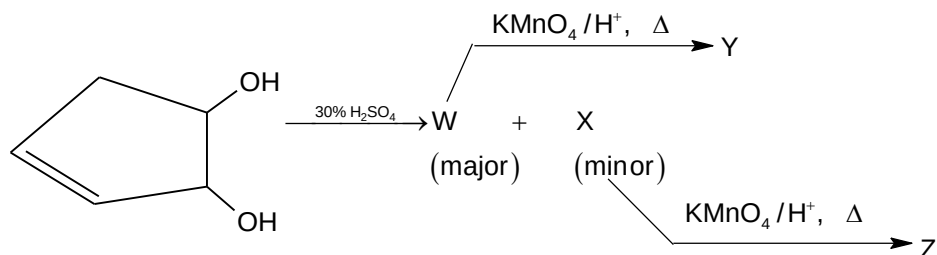
- (A) E₁ (B) E_{1cb}
(C) E₂ (D) none
-

Space for rough work

PART – A: (Multiple Correct Answers Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

5.

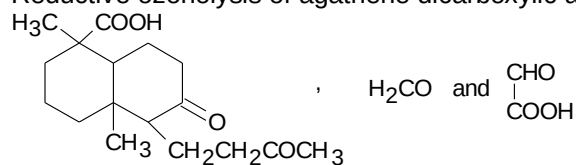


Of the 4 compounds listed above, more than one will:

- (A) exhibit resonance due to conjugation in their structure
- (B) show NaHCO_3 test
- (C) have a cyclic structure
- (D) show 2,4-DNPH precipitation

6.

Reductive ozonolysis of agathene dicarboxylic acid gives



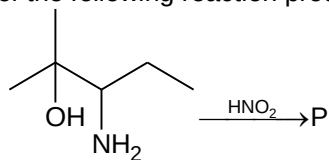
Which of the following structure(s) can not be of agathene dicarboxylic

- (A)
- (B)
- (C)
- (D)

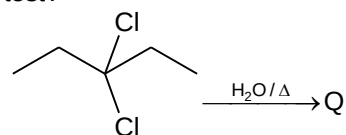
Space for rough work

7. Which of the following reaction products give haloform test?

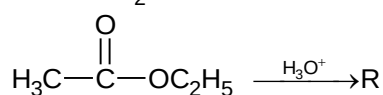
(A)



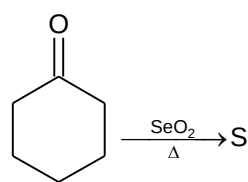
(B)



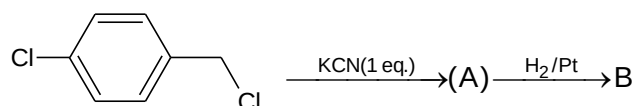
(C)



(D)



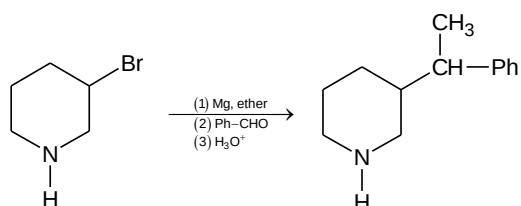
8.



Correct statement for product B is

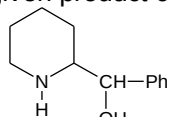
- (A) It gives positive carbyl amine test
- (B) It gives coupling reaction with diazonium salt
- (C) It gives alcohol on treatment with nitrous acid
- (D) It forms dibromo derivative with bromine water

9.

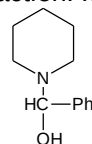


The given product cannot be obtained in the above reaction. Identify the correct product obtained.

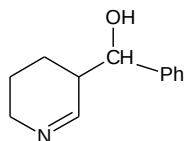
(A)



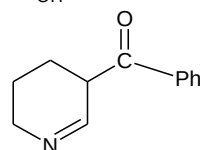
(B)



(C)

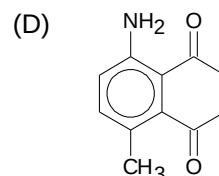
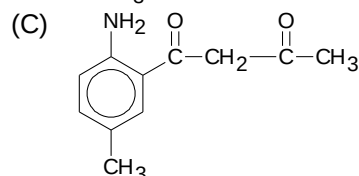
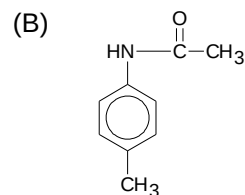
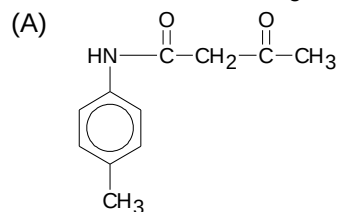
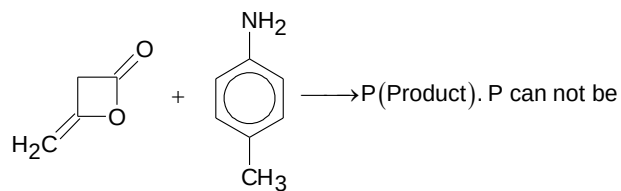


(D)



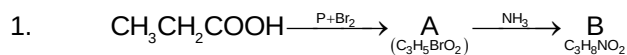
Space for rough work

10.



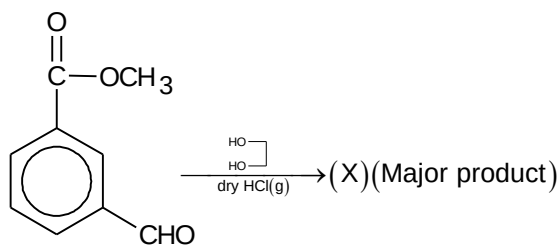
PART – C: (Numerical Based Type)

This section contains **08 questions**. The answer to each question is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the ORS is to be bubbled.



If dl ($\pm$) mixing of (B) heated then the possible. Number of stereo isomer formed are.....

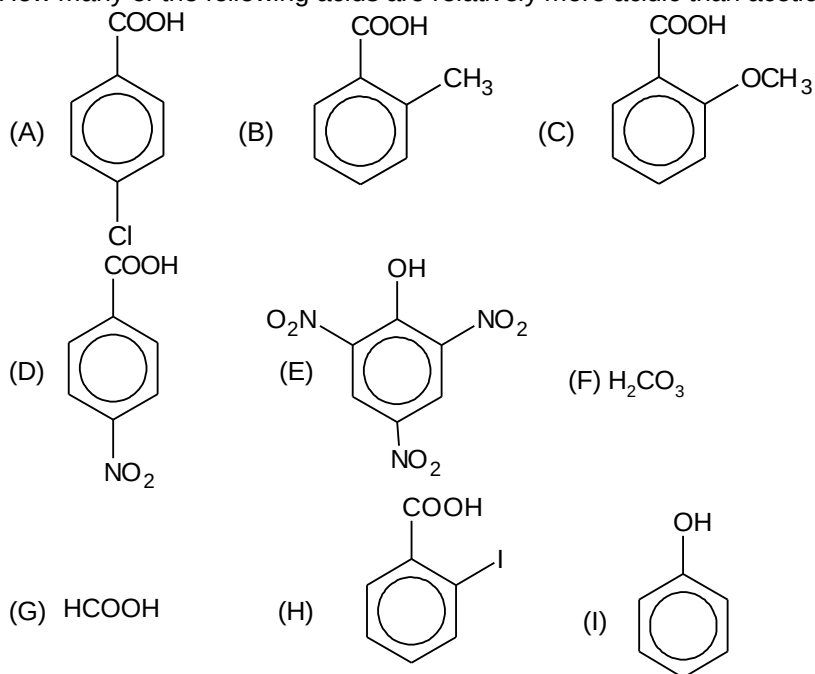
2.



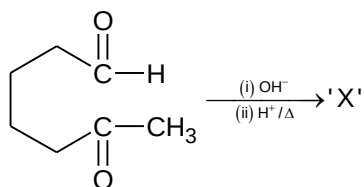
Degree of unsaturation of 'X' is.....

Space for rough work

3. How many of the following acids are relatively more acidic than acetic acid?



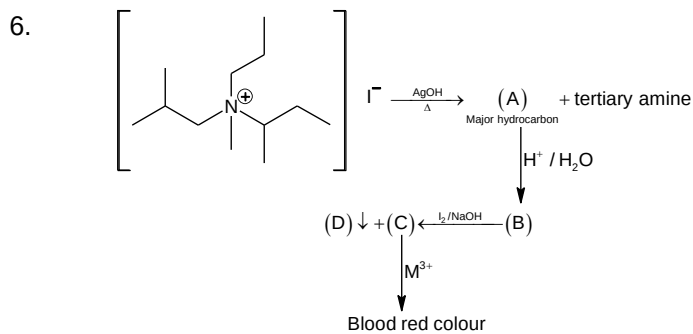
4.



Find degree of unsaturation of 'X'.

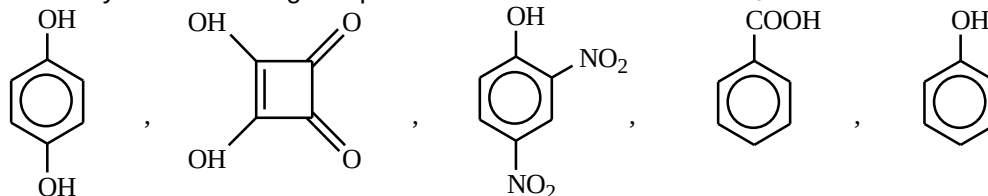
5. When $\text{H}-\underset{\text{D}}{\overset{\text{Ph}}{\text{C}}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$, $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$ and $\text{H}-\underset{\text{D}}{\overset{\text{CH}_3}{\text{C}}}-\text{CONH}_2$ are mixed and reacted with Br_2/KOH than how many 1° amines are obtained.

Space for rough work

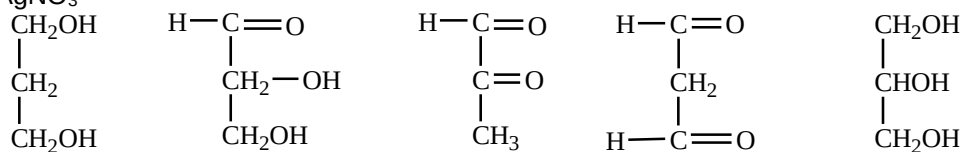


The number of electrons in d-orbitals of M^{3+} is (where atomic number of $\text{M} \leq 30$).

7. How many of the following compounds liberate CO_2 with NaHCO_3



8. How many of the following compounds give white precipitate when treated with HIO_4 followed by AgNO_3



Space for rough work